Remote
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Nathaniel Vazquez

SOFTWARE ENGINEER

SUMMARY

Software engineer with a foundation in semiconductor manufacturing, enterprise operations, and hands-on education — now building at the intersection of quantum computing and applied cryptography. Founder of Quantum V^ LLC, where I independently designed and shipped STEADYWATCH: an iOS/watchOS biometric identity platform integrating IBM quantum hardware entropy, Apple Secure Enclave P256 signing, and atomic clock-precision biometric capture. Comfortable operating across the full stack — from low-level hardware and sensor data to backend API architecture and polished mobile UI. Background in data-driven operations, team leadership, and technical instruction informs an engineering style that is systematic, well-documented, and built to last.

EXPERIENCE

Founder & Lead Software Engineer

2025 — Present

Quantum V^ LLC · Remote

- **Designed and shipped STEADYWATCH** — a dual-platform iOS/watchOS app implementing an atomic biometric identity pipeline: biological oscillation frequency ÷ Cesium-133 atomic clock standard → IBM quantum entropy seeding → Secure Enclave P256 ECDSA signing.
- Built and maintains a Python/Flask quantum computing backend (75+ REST endpoints) integrating IBM Quantum hardware (ibm_marrakesh, ibm_fez, ibm_kingston) and local AerSimulator; developed Echo Resonance™ post-quantum encryption architecture with a 2^{4096} key space, resistant to Shor's algorithm.
- Conducted original quantum hardware research: Hurwitz-Mermin GHZ experiment validated Bell inequality violations across three IBM Quantum processors (M = 3.82, 95.6% of quantum maximum on ibm_marrakesh); paper submitted to peer-reviewed Quantum Journal.
- Filed provisional patent covering biometric key derivation from oscillation frequency, quantum entropy seeding, and post-quantum encryption architecture; NSF SBIR application submitted.
- Managed full product lifecycle: Swift/SwiftUI development, Python backend, IBM quantum job orchestration, App Store submission, privacy compliance, and business development.
- **Designed and shipped KaizenDoku**, a React Native/Expo iOS app on the App Store, architecting a local-first storage layer (MMKV/AsyncStorage) that replaced a cloud backend to remove network dependency from core gameplay.
- Built real-time multiplayer (Socket.IO room/lobby management, turn-based state sync for Versus and Co-op modes) backed by a self-hosted Node.js/Express + SQLite server deployed behind a Cloudflare Tunnel for board distribution and leaderboard submission.
- Integrated Apple Game Center (leaderboards, achievements), HealthKit, and Apple Watch heart-rate capture; implemented AdMob monetization (banner/interstitial/rewarded placements) alongside a premium unlock tier.

Senior Software Developer & Teaching Assistant

2022 — 2025

Persevere NOW · Remote

- Served as Senior Software Developer and full-time Teaching Assistant for multiple cohorts of 20 students, teaching software engineering fundamentals to underserved communities.
- Designed and delivered curriculum covering web development, programming fundamentals, and software engineering best practices; adapted instruction to diverse learning backgrounds and skill levels.
- Mentored students through project development from concept to deployment; provided code reviews, debugging support, and career guidance for aspiring developers entering the workforce.

Lead Pharmacy Technician — DOD Overseas & Expatriate Team

2012 — 2018

Express Scripts · Remote / Enterprise

- Led a team of ~20 CPhT-licensed professionals providing pharmaceutical services to deployed Department of Defense personnel and expatriate staff overseas; managed scheduling, quality control, and performance development.
- Drove data collection and analysis workflows to monitor prescription throughput, error rates, and compliance metrics across a high-volume mail-order fulfillment operation.
- Role evolved into enterprise-level management covering the full scope of overseas mail-order pharmaceutical services — coordinating logistics, regulatory compliance, and cross-functional team operations at scale.

Process Engineer — Test Wafer Distribution

2005 — 2011

Freescale Semiconductor · Remote / On-site

- Operated, tested, and maintained CVD, Cleans, Diffusion, EPI, and Photolithography equipment in the design and manufacture of microprocessors within a semiconductor fabrication environment.
- Role specialized into the management, data collection/analysis, and deployment of test silicon across the entirety of the manufacturing process — tracking wafer performance metrics and identifying process deviations.
- Developed hands-on expertise in precision manufacturing, process control, and hardware-level data analysis that directly informs current work in quantum hardware and cryptographic systems.

TECHNICAL SKILLS

LANGUAGES

Swift · SwiftUI · Python · JavaScript · HTML/CSS

MOBILE

iOS · watchOS · React Native · Expo · CoreMotion · HealthKit · WatchConnectivity · LocalAuthentication · CryptoKit · Secure Enclave

BACKEND

Python · Flask · Node.js · Express · Socket.IO · REST API design · JSON · SQLite

QUANTUM COMPUTING

Qiskit · IBM Quantum hardware · AerSimulator · GHZ circuits · QKD protocols · Mermin inequality

CRYPTOGRAPHY

P256 ECDSA · SHA-256 · Post-quantum encryption · Secure Enclave key management · Biometric key derivation

TOOLS & PRACTICES

Xcode · Git · GitHub · LaTeX · Agile · Technical writing · Patent documentation

NOTABLE WORK

IBM Quantum Hardware Validation

Designed and executed Hurwitz-Mermin GHZ experiments across three IBM Quantum processors. Achieved M = 3.82 (95.6% of quantum maximum) on ibm_marrakesh. All results reproducible via public IBM job IDs.

STEADYWATCH — App Store v1.0.3

End-to-end iOS + watchOS app: biometric key generation, IBM quantum entropy vault (81 slots), Face ID + Secure Enclave authorization pipeline, real-time quaternion visualization.

Peer-Reviewed Research Submission

"Bell Inequality Violations from Hurwitz Quaternion Lattice Geometry" — submitted to Quantum Journal. Documents self-similarity discovery across F4 lattice prime shells.

Patent Pending · NSF SBIR Applicant

Provisional patent filed covering biometric oscillation key derivation and Echo Resonance post-quantum architecture. NSF SBIR application submitted for quantum security commercialization.

EDUCATION & CERTIFICATIONS

Certified Pharmacy Technician (CPhT)

Active during tenure at Express Scripts

Pharmacy Technician Certification Board (PTCB)

Semiconductor Process Engineering

2005 — 2011

On-the-job training · Freescale Semiconductor · CVD, EPI, Diffusion, Photolithography